



UNIVERSITY OF
CAMBRIDGE

Department of Computer
Science and Technology

Accelerating Molecular Generation through Representation Alignment

Shreyas Ravishankar

Hughes Hall

June 2026

Submitted in partial fulfillment of the requirements for the
Master of Philosophy in Advanced Computer Science

Total page count: 23

Main chapters (excluding front-matter, references and appendix): 13 pages (pp 6–18)

Main chapters word count: 467

Methodology used to generate that word count:

[For example:

```
$ make wordcount
gs -q -dSAFER -sDEVICE=txtwrite -o - \
    -dFirstPage=6 -dLastPage=11 report-submission.pdf | \
egrep '[A-Za-z]{3}' | wc -w
467
```

]

Declaration

I, Shreyas Ravishankar of Hughes Hall, being a candidate for the Master of Philosophy in Advanced Computer Science, hereby declare that this report and the work described in it are my own work, unaided except as may be specified below, and that the report does not contain material that has already been used to any substantial extent for a comparable purpose. In preparation of this report, I adhered to the Department of Computer Science and Technology AI Policy. [The project required the approved use of {insert name of technology} and such use is acknowledged in the text at the relevant sections.] [I am content for my report to be made available to the students and staff of the University.]

Signed:

Date:

Abstract

Write a summary of the whole thing. Make sure it fits on one page.

Contents

1	Introduction	6
2	Background	9
2.1	Flow matching	9
2.2	Representation alignment	10
3	Related work	13
4	Evaluating molecular generation	14
5	Encoder targets and profiling	15
6	REPA on small molecules	16
7	REPA on protein backbones	17
8	Conclusions	18
A	Technical details	23
A.1	Placeholder section	23

Chapter 1

Introduction

The evolution of generative machine learning in scientific domains is defined by the tension between data scale, data quality, and structural priors. Our report examines a recent addition to this story in *representation alignment*. We study this technique in two regimes for de novo molecular generation — small molecules and protein backbones — and characterise its behaviour. In this introduction, we trace the narrative arc of research trends in molecular generation, and highlight the motivation for our work: the emerging need for data-efficient modelling in a world of prior-light models.

The dominant ideas in generative modelling are *denoising diffusion* and its offshoot *flow matching* [1, 2], which found breakthrough success in the realm of image generation. This paradigm has since been adapted for de novo design challenges across scientific fields, such as generating protein backbones [17, 18, 13], inorganic materials [20], and drug-like small molecules [14]. Training these models is typically expensive in both data and compute, with state-of-the-art image generators routinely training on billions of examples [27]. However, the corpus of available high-quality molecular data sits orders of magnitude below this. Experimental determination is slow and expensive, which means datasets are correspondingly small. For example, protein structure generators have largely been trained on at most half a million structures [13], which may explain why they have not yet matched the maturity of their vision-domain counterparts.

Historically, researchers compensated for the small scale of high-quality molecular data by encoding structural priors directly into model architectures [33]. SE(3)-equivariant architectures, for example, enforce the rotational and translational symmetries expected of molecules by construction [35]. But hard structural priors come with a computational cost, and the operations required to encode geometry are a bottleneck to scaling models.

A recent wave of research has moved entirely the other way: drop hard priors, use bare-bones non-equivariant transformer stacks, and rely on data scale and augmentation to teach the model what equivariance buys. This shift fits a broader trend in machine learning on leveraging representation instead of model architecture to inject soft inductive

biases for structured data. Vision Transformers [36] showed that the spatial-locality and translation-equivariance priors of convolutional neural nets (CNNs) can be learned rather than imposed, given sufficient scale. Graph Transformers like TokenGT [37] have made the same case for graphs. AlphaFold 3 [39] stripped many of the SE(3)-equivariant biases of AlphaFold 2 [38], replacing the structure module with a diffusion model operating on raw coordinates. Brehmer et al. [40] make the empirical scaling case directly: more data through a simpler model beats less data through a richer one. In our work, we examine Tabasco [14] and Proteina [13], which are flow-based generators built on vanilla transformers for small molecules and protein backbones respectively.

Given the difficulty of procuring data in scientific domains, data scale must necessarily come from non-experimental sources, which may be of lower quality. For example, recent efforts in protein generation have leaned on synthetic structures from folding-model predictions [13, 29] to expand training datasets into the tens of millions. However, even these enhanced corpora are arguably insufficient. Metagenomic databases catalogue over two billion natural protein sequences [31], and the theoretical sequence space at a median protein length [32] is hundreds of orders of magnitude greater still. Even at the frontier of the field’s data scale, we sample but a sliver of the distribution we wish to model.

In summary, the field has traded hard structural priors for architectural simplicity, and curated data for synthetic scale. However, this trade-off dilutes the structural signal available to a model. Furthermore, even our best attempts at creating large datasets may not be large enough. These twin challenges amplify the need for techniques that make training still more efficient without sacrificing quality.

But why is training diffusion-transformer architectures expensive in the first place? One answer comes from Yu et al. [5], who argue that these models suffer from a *representation bottleneck*. During training, a model is implicitly asked to learn two distinct tasks simultaneously: (1) representation: extracting semantically meaningful encodings from complex data; and (2) generation: constructing data from these encodings. The standard denoising objective alone may not provide sufficient signal for representation learning, leading to slow convergence and the large training budgets these models require in practice. In the context of molecular generation, this representation burden is compounded by the twin challenges outlined above. Models are now forced to encode complex structural geometry from data that is vastly smaller than the true underlying distribution, all while operating without the aid of architectural priors.

Yu et al. propose Representation Alignment (REPA) as a remedy. The technique adds a regularisation term that aligns a generator’s intermediate hidden states with the embeddings of a frozen pretrained encoder. On image diffusion, they report roughly a $17.5\times$ training speed-up against a strong baseline. REPA has since been extended beyond images to video [9], audio [10], and recently to molecular generation [7, 8]. However, whether REPA’s gains transfer to molecular settings, and why, remain uncharacterised.

In this light, the core question we investigate in this report is: *can prior-light molecular generation models benefit from representation alignment with pre-trained models, in training efficiency and generation quality?* We aim to characterise when and why this works, including failure modes and limitations.

Our report documents two lines of enquiry.

- (i) *Does REPA work in this domain?*: We conduct empirical studies on training flow-based molecular generator architectures with REPA in two regimes: Tabasco [14] for small molecules, and Proteina [13] for protein backbones. We find that REPA does not measurably improve Tabasco, but it improves training efficiency and advances the generation quality envelope for Proteina.
- (ii) *When does REPA work?*: Following Singh et al. [6] in the realm of image generation, we develop a profiling framework to characterise the suitability of an encoder as a representation target. We measure three properties of an encoder: (1) an upper bound on the structural information it contains (linear-probe recoverability against domain structural labels); (2) a lower bound on what a projector can transfer from that information (projector-based recoverability); (3) the geometric conditioning of its embeddings for the alignment objective (effective rank).

We do not propose a new architecture, alignment loss, or evaluation metric, and our two studies are not a controlled experiment over the conditions that govern REPA’s effectiveness. Instead, they are best read as paired case studies in regimes where those conditions differ.

The remainder of this report is organised as follows. Chapter 2 sets out the background needed to read the rest: flow matching, REPA, and the parameterisations we use for small molecules and proteins. Chapter 3 places the work in context, covering the scaling-over-priors trend in structural biology and the recent literature on representation alignment. Chapter 4 discusses what it means for a generative model to be “good” in our domain, and the tensions that question hides. Chapter 5 develops the encoder-profiling framework and uses it to predict where REPA may and may not help. Chapters 6 and 7 present the two paired studies, small molecules first and protein backbones second. Finally, we conclude in Chapter 8 with a discussion of whether what we learn generalises beyond molecular generation.

Chapter 2

Background

In this chapter, we introduce the machinery the rest of our report depends on. We outline flow matching (§2.1), the generative paradigm underlying the models we study, and representation alignment (§2.2), the regulariser that is our object of study. We then provide some motivating context on small molecules and protein backbones (§2.3), and discuss how they are typically parameterised for generative machine learning workflows. Finally, we outline Tabasco and Proteina (§2.4), the two flow-matching backbones we extend in our experiments. The treatment is cursory by design, intended only to guide the reader through this report, and we provide citations for omitted details.

2.1 Flow matching

At first glance, the term *generative modelling* implies the problem of *creating* new artefacts from scratch. However, it is perhaps more accurately described as the problem of *sampling* new artefacts from a complex data distribution that can only be accessed through a finite training set. A common strategy here is to fix a simple distribution we can sample from cheaply — typically standard Gaussian noise $p_0 = \mathcal{N}(0, I)$ — and learn a mapping from it to the desired data distribution p_1 .

Flow matching realises this mapping as a continuous-time transport. The model designer defines a *probability path* between a noise sample and a data sample — a continuous interpolation between them — and the model learns to carry the noise sample along this path. Concretely, it parameterises a time-dependent *velocity field* $v_\theta(x, t)$ that gives the instantaneous direction of motion at each location x and time $t \in [0, 1]$. At inference, integrating the learned ordinary differential equation (ODE) $\dot{x}_t = v_\theta(x_t, t)$ from $t = 0$ to $t = 1$ pushes a fresh noise sample forward into a fresh data sample.

A common choice of probability path is the linear interpolant, $x_t = (1 - t)x_0 + tx_1$ for $x_0 \sim p_0$ and $x_1 \sim p_1$. The model’s target is the *marginal* velocity field at (x, t) : the average direction of motion over all noise-data pairs whose paths pass through that point.

However, supervising this marginal directly is intractable, since the contributing pairs are not known in advance. Flow matching sidesteps this difficulty with a key observation [2]: although the marginal field may be intractable, the *conditional* velocity along the path of any specific noise-data pair is relatively trivial. Indeed, for the linear interpolant, it is simply $\dot{x}_t = x_1 - x_0$. Regressing the model on these conditional velocities, averaged over many sampled pairs, recovers the marginal field in expectation, and gives the training loss as a mean-squared regression:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, x_0, x_1} \|v_\theta(x_t, t) - (x_1 - x_0)\|^2.$$

In practice, every training step samples a time $t \in [0, 1]$ and a noise-data pair (x_0, x_1) , computes the interpolant x_t , and takes a gradient step on this squared error.

Flow matching with this linear interpolant generalises score-based diffusion: the variance-preserving Gaussian-path objective of denoising diffusion models [1] is a special case of the FM family [2, 4].

Generating samples from a trained flow-matching model amounts to integrating the learned velocity field. This is often performed with a standard ODE solver, with Euler steps usually sufficing for stable trajectories. The field also admits stochastic sampling variants — such as those drawing on the connection to score-based stochastic differential equations (SDEs) — which inject noise during the integration steps to explore the probability space and improve sample diversity. The models we work with in this report, Tabasco and Proteina, support both deterministic and stochastic samplers.

Conditional generation augments v_θ with context tokens — such as a class label, a sequence length, or a structural motif — with conditioning supplied at both training and sampling time. In this report, we focus on unconditional generation only.

For a more complete treatment of flow matching, we refer the reader to Lipman et al. [2] for the original presentation, and to the recent Lipman et al. guide [4].

2.2 Representation alignment

Chapter 1 introduced the argument made by Yu et al. [5] about the *representation bottleneck* in diffusion-transformer models. Their analysis applies to flow-matching models as well, given the equivalence between the two training objectives discussed above (§2.1). The flow-matching objective also implicitly asks a model to learn two things at once: a useful representation of the data, and a noise-to-data velocity field that interprets this representation. This twin burden, they argue, contributes to slow convergence and large training budgets.

Their proposed solution, Representation Alignment (REPA), addresses the first half of this burden by adding a regularisation term to the training loss. Figure 2.1 shows the

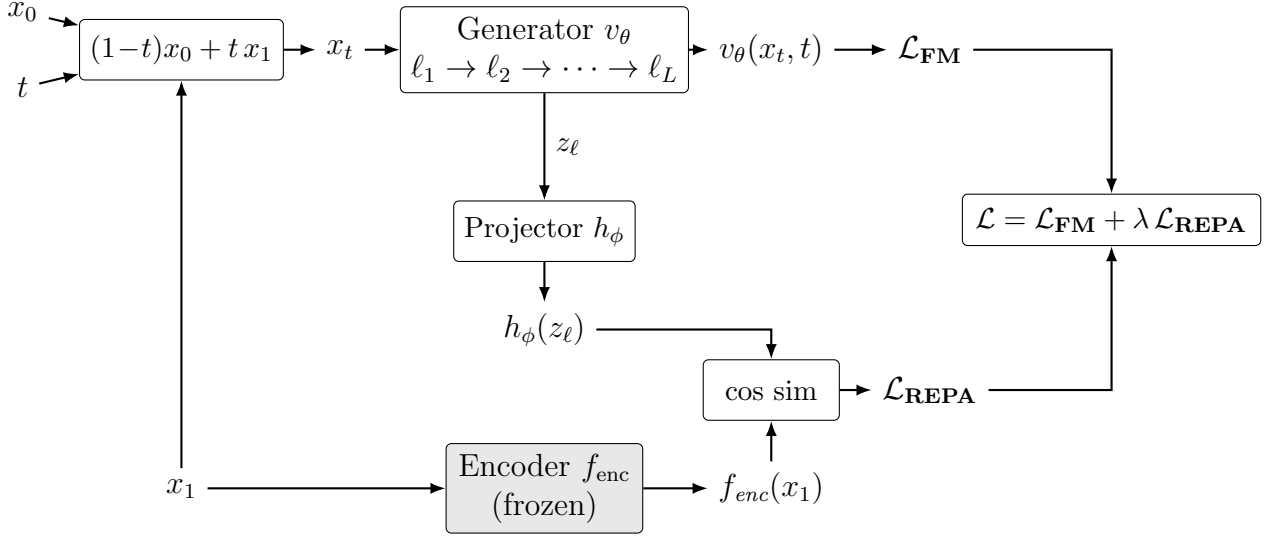


Figure 2.1: The REPA construction. A noised sample $x_t = (1-t)x_0 + tx_1$ is constructed at each training step from a noise sample x_0 , a clean data sample x_1 , and a sampled timestep $t \in [0, 1]$. The generator v_θ processes x_t through its sequence of layers $\ell_1, \dots, \ell_L$ to produce the velocity field that feeds the flow-matching loss $\mathcal{L}_{\text{FM}}$. Separately, the clean sample x_1 is also passed through a frozen pretrained encoder f_{enc} (shaded). The generator’s hidden state z_ℓ at the chosen alignment layer is mapped into the encoder’s feature space by a trainable projector h_ϕ ; the REPA loss $\mathcal{L}_{\text{REPA}}$ is the negated cosine similarity between $h_\phi(z_\ell)$ and $f_{\text{enc}}(x_1)$, averaged across tokens. The total training loss $\mathcal{L}$ combines them with weight λ .

construction. The regularisation pulls z_ℓ , the generator’s intermediate hidden state at a chosen *alignment layer* ℓ , towards the embeddings produced by a frozen pretrained encoder f_{enc} run on the clean data sample x_1 . z_ℓ is a sequence of N token vectors, corresponding to image patches in the original setting, and atoms or residues in the molecular domains we study (§2.3). A small trainable *projector head* h_ϕ — a three-layer multi-layer perceptron (MLP) in the original paper — maps z_ℓ into the encoder’s feature space, so that $h_\phi(z_\ell)$ and $f_{\text{enc}}(x_1)$ can be compared. Crucially, this alignment is asymmetric: gradients update only the generator parameters θ and the projector ϕ , never the encoder. Information thus flows one way, from the pretrained encoder into the generator’s hidden representations.

We follow the original paper in using cosine similarity — Yu et al. compared cosine and mean-squared error for image diffusion and found cosine more effective, a result we corroborate in the molecular setting (Chapter 7). The alignment loss is then the negated cosine similarity between projected and target embeddings, averaged jointly over training samples and tokens:

$$\mathcal{L}_{\text{REPA}} = -\mathbb{E}_{t, x_0, x_1} \left[\frac{1}{N} \sum_{n=1}^N \cos(h_\phi(z_\ell)_n, f_{\text{enc}}(x_1)_n) \right].$$

In the image setting N is fixed across the dataset, so this inner average is equivalent

to averaging across all tokens in the minibatch. For molecular data, where N varies per sample, the two diverge: averaging tokens within each sample first weights every molecule equally, while pooling tokens across the batch gives larger molecules proportionally more influence. We return to this distinction in the design knobs below.

The final training objective is the flow-matching loss from §2.1 plus this term, scaled by a weight λ :

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda \mathcal{L}_{\text{REPA}}.$$

This construction exposes several design knobs, whose effects we revisit in the ablations of Chapter 7:

- The choice of encoder f_{enc} , as different encoders may capture distinct aspects of the data distribution.
- The alignment layer ℓ at which we attach the projector. The right choice is not obvious a priori. The original REPA paper experimentally determined that mid-network layers work best for image diffusion, and we investigate this in the context of molecular generators.
- The alignment-loss weight λ , which balances the alignment signal against the generative objective.
- The averaging mode. The equation above is the *per-sample* form: cosine similarities are averaged within each sample (the $1/N$ factor) before the expectation across samples. The alternative *per-token* form weights each token equally across the batch, giving larger molecules more influence.

In this report, we adopt REPA in its original form: cosine similarity, additive combination with the flow-matching loss, and a frozen encoder. A growing family of variants relaxes one or more of these choices — for example, encoder-free dispersive regularisers [12] that drop the external encoder entirely, or joint-training extensions that fine-tune the encoder alongside the generator. We defer their discussion to Chapter 3.

Chapter 3

Related work

Chapter 4

Evaluating molecular generation

Chapter 5

Encoder targets and profiling

Chapter 6

REPA on small molecules

Chapter 7

REPA on protein backbones

Chapter 8

Conclusions

Bibliography

- [1] Jonathan Ho, Ajay Jain, Pieter Abbeel. *Denoising Diffusion Probabilistic Models*. NeurIPS 2020. arXiv:2006.11239.
- [2] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, Matt Le. *Flow Matching for Generative Modeling*. ICLR 2023. arXiv:2210.02747.
- [3] Peter Holderrieth and Ezra Erives. *An Introduction to Flow Matching and Diffusion Models*. MIT 6.S184 lecture notes, 2025. diffusion.csail.mit.edu.
- [4] Yaron Lipman, Marton Havasi, Peter Holderrieth, Neta Shaul, Matt Le, Brian Karrer, Ricky T. Q. Chen, David Lopez-Paz, Heli Ben-Hamu, Itai Gat. *Flow Matching Guide and Code*. 2024. arXiv:2412.06264.
- [5] Sihyun Yu, Sangkyung Kwak, Huiwon Jang, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, Saining Xie. *Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think*. ICLR 2025. arXiv:2410.06940.
- [6] Jaskirat Singh, Xingjian Leng, Zongze Wu, Liang Zheng, Richard Zhang, Eli Shechtman, Saining Xie. *What matters for Representation Alignment: Global Information or Spatial Structure?* 2025. arXiv:2512.10794.
- [7] Chenyu Wang, Cai Zhou, Sharut Gupta, Zongyu Lin, Stefanie Jegelka, Stephen Bates, Tommi Jaakkola. *Learning Diffusion Models with Flexible Representation Guidance*. 2025. arXiv:2507.08980.
- [8] Hyosoon Jang, Hyunjin Seo, Yunhui Jang, Seonghyun Park, Sungsoo Ahn. *Boltz is a Strong Baseline for Atom-level Representation Learning*. 2026. arXiv:2602.13249.
- [9] Xiangdong Zhang, Jiaqi Liao, Shaofeng Zhang, Fanqing Meng, Xiangpeng Wan, Junchi Yan, Yu Cheng. *VideoREPA: Learning Physics for Video Generation through Relational Alignment with Foundation Models*. 2025. arXiv:2505.23656.
- [10] Tri Ton, Jiajing Hong, Chang D. Yoo. *Temporally Aligned Audio for Video with Autoregression*. ICCV 2025. arXiv:2504.05684.
- [11] Wei Zhang, Shijie Jin, Hangrui Zhang, Yi Li, Yan Wang. *CrystalREPA: Element-aware Representation Alignment for Crystal Generation*. 2026. arXiv:2605.08960.

- [12] Runqian Wang and Kaiming He. *Diffuse and Disperse: Image Generation with Representation Regularization*. 2025. arXiv:2506.09027.
- [13] Tomas Geffner et al. *Proteina: Scaling Flow-based Protein Structure Generative Models*. ICLR 2025. arXiv:2503.00710.
- [14] C. Vonessen et al. *TABASCO: A Fast, Simplified Model for Molecular Generation with Improved Physical Quality*. 2025. arXiv:2507.00899.
- [15] Marcus Olivecrona, Thomas Blaschke, Ola Engkvist, Hongming Chen. “Molecular de-novo design through deep reinforcement learning”. *Journal of Cheminformatics* 9:48 (2017). doi:10.1186/s13321-017-0235-x.
- [16] Alex Zhavoronkov et al. “Deep learning enables rapid identification of potent DDR1 kinase inhibitors”. *Nature Biotechnology* 37 (2019), pp. 1038–1040. doi:10.1038/s41587-019-0224-x.
- [17] Joseph L. Watson et al. “De novo design of protein structure and function with RFDiffusion”. *Nature* 620 (2023), pp. 1089–1100. doi:10.1038/s41586-023-06415-8.
- [18] John B. Ingraham et al. “Illuminating protein space with a programmable generative model”. *Nature* 623 (2023), pp. 1070–1078. doi:10.1038/s41586-023-06728-8.
- [19] Amil Merchant et al. “Scaling deep learning for materials discovery”. *Nature* 624 (2023), pp. 80–85. doi:10.1038/s41586-023-06735-9.
- [20] Claudio Zeni et al. “A generative model for inorganic materials design”. *Nature* 639 (2025), pp. 624–632. doi:10.1038/s41586-025-08628-5.
- [21] Liang Wang, Chao Song, Zhiyuan Liu, Yu Rong, Qiang Liu, Shu Wu, Liang Wang. *Diffusion Models for Molecules: A Survey of Methods and Tasks*. 2025. arXiv:2502.09511.
- [22] Zeming Lin et al. “Evolutionary-scale prediction of atomic-level protein structure with a language model”. *Science* 379.6637 (2023), pp. 1123–1130. doi:10.1126/science.ade2574.
- [23] J. Dauparas et al. “Robust deep learning-based protein sequence design using ProteinMPNN”. *Science* 378.6615 (2022), pp. 49–56. doi:10.1126/science.add2187.
- [24] Ilyes Batatia et al. “A foundation model for atomistic materials chemistry”. *The Journal of Chemical Physics* 163.18 (2024), 184110. doi:10.1063/5.0155322.
- [25] Michael Heinzinger et al. “Bilingual language model for protein sequence and structure”. *NAR Genomics and Bioinformatics* 6.4 (2024), lqae150. doi:10.1093/nargab/lqae150.
- [26] Jin Su et al. “SaProt: Protein Language Modeling with Structure-aware Vocabulary”. *bioRxiv* (2023). Preprint. doi:10.1101/2023.10.01.560349.

- [27] Christoph Schuhmann et al. *LAION-5B: An open large-scale dataset for training next generation image-text models*. NeurIPS 2022 Datasets and Benchmarks. arXiv:2210.08402.
- [28] Helen M. Berman et al. “The Protein Data Bank”. *Nucleic Acids Research* 28.1 (2000), pp. 235–242. Statistics retrieved Nov 2025, rcsb.org/stats.
- [29] Mihaly Varadi et al. “AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences”. *Nucleic Acids Research* 52 (2024), D368–D375. doi:10.1093/nar/gkad1011.
- [30] Simon Axelrod and Rafael Gómez-Bombarelli. “GEOM, energy-annotated molecular conformations for property prediction and molecular generation”. *Scientific Data* 9, 185 (2022). doi:10.1038/s41597-022-01288-4.
- [31] Lorna Richardson et al. “MGnify: the microbiome sequence data analysis resource in 2023”. *Nucleic Acids Research* 51 (2023), D753–D759. doi:10.1093/nar/gkac1080. Release statistics retrieved May 2022 (2.4 billion non-redundant sequences): ebi.ac.uk.
- [32] Luciano Brocchieri and Samuel Karlin. “Protein length in eukaryotic and prokaryotic proteomes”. *Nucleic Acids Research* 33.10 (2005), pp. 3390–3400. doi:10.1093/nar/gki615.
- [33] Michael M. Bronstein, Joan Bruna, Taco Cohen, Petar Veličković. *Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges*. 2021. arXiv:2104.13478.
- [34] Alexandre Duval, Simon V. Mathis, Chaitanya K. Joshi, Victor Schmidt, Santiago Miret, Fragkiskos D. Malliaros, Taco Cohen, Pietro Liò, Yoshua Bengio, Michael Bronstein. *A Hitchhiker’s Guide to Geometric GNNs for 3D Atomic Systems*. 2024. arXiv:2312.07511.
- [35] Nathaniel Thomas, Tess Smidt, Steven Kearnes, Lusann Yang, Li Li, Kai Kohlhoff, Patrick Riley. *Tensor Field Networks: Rotation- and Translation-Equivariant Neural Networks for 3D Point Clouds*. 2018. arXiv:1802.08219.
- [36] Alexey Dosovitskiy et al. *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. ICLR 2021. arXiv:2010.11929.
- [37] Jinwoo Kim, Tien Dat Nguyen, Seonwoo Min, Sungjun Cho, Moontae Lee, Honglak Lee, Seunghoon Hong. *Pure Transformers are Powerful Graph Learners*. NeurIPS 2022. arXiv:2207.02505.
- [38] John Jumper et al. “Highly accurate protein structure prediction with AlphaFold”. *Nature* 596 (2021), pp. 583–589. doi:10.1038/s41586-021-03819-2.
- [39] Josh Abramson et al. “Accurate structure prediction of biomolecular interactions with AlphaFold 3”. *Nature* 630 (2024), pp. 493–500. doi:10.1038/s41586-024-07487-w.

- [40] Johann Brehmer et al. *Does equivariance matter at scale?* NeurIPS 2024. arXiv:2410.23179.

Appendix A

Technical details

A.1 Placeholder section

TODO.